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Introduction

More than 100 distinct types of cancer have been identified. Cancer frequently affects multiple organs due to uncontrolled cell division, migration, and invasion. Understanding its pathogenesis is crucial, given the rising number of cases and the limitations of standard treatment options, which contribute to poor overall and disease-free survival rates for many patients.

The somatic mutation theory became widely accepted as the primary cause of cancer. However, certain exceptions highlight the need to consider all entities of molecular interactions (EoMIs) in understanding carcinogenesis. EoMI includes various processes such as synthetic, mechanical, electrical, concentration, and heat generation-related interactions, all of which should be factored into cancer research.

Currently, the most widely accepted theory is the multistep theory, which explains the genetic and epigenetic events involved in transforming a single cell into numerous malignant cells capable of metastasis. Moreover, identifying cancer hallmarks has redefined the understanding of carcinogenesis, providing a more comprehensive view of cancer as a process driven by the accumulation of these hallmarks

Causes

Cancer usually develops through the transformation of normal cells into malignant tumor cells, often progressing from a precancerous condition to a fully developed tumor. This transition is influenced by an individual's genetics and exposure to 3 main categories, as mentioned below.

Physical Carcinogens

- Ultraviolet and ionizing radiation

Chemical Carcinogens

- Polycyclic hydrocarbons
- Asbestos
- Components of tobacco smoke
- Alcohol
- Aflatoxin (a grain contaminant)
- Arsenic (a drinking water contaminant)

Biological Carcinogens

- Viral infections such as hepatitis B and human papillomavirus (HPV)
- Bacterial infections, such as *Helicobacter pylori*
- Parasitic infections, such as *Schistosoma haematobium*

Biochemical and Genetic Pathology

Cancer is a group of diseases characterized by diverse molecular signatures, yet all share common cellular phenotypic traits and functional capabilities. Hanahan and Weinberg identified these traits as the hallmarks of cancer, which include increased cell proliferation, achieved through sustained proliferative signaling or evasion of growth suppressors. Additionally, cancer cells promote tumor progression and spread by inducing or accessing vasculature and activating invasion and metastasis.

Two enabling characteristics facilitate the acquisition of these traits: genomic instability and tumor-promoting inflammation. Additionally, deregulated tumor metabolism has emerged as a defining feature of cancer cells. Carcinogenesis can be viewed as a series of genetic and epigenetic changes that gradually allow cells to acquire these hallmarks, ultimately leading to full cancer development.

The carcinogenesis process involves multiple events of genetic DNA damage (mutations) and epigenetic modifications in gene expression that accumulate sequentially. These changes contribute to the gradual acquisition of the hallmarks of cancer, ultimately resulting in fully developed, metastatic cancer.

DNA Damage

DNA damage plays a crucial role in carcinogenesis. Proper cell function depends on the precise regulation of various cellular processes, including DNA replication and repair. When DNA is damaged, it typically activates repair mechanisms. As DNA serves as the genetic blueprint for all living organisms, damage can significantly impact the cell and its future progeny. To counteract this, cells have highly efficient DNA repair systems designed to prevent harmful mutations.

If DNA damage is irreparable, the affected cells undergo programmed cell death, also known as apoptosis. When apoptosis fails, a third line of defense—the immune system—is activated to recognize and attack cells with mutated genes. However, if the immune system is compromised or overwhelmed by the mutated cells, these cells can persist and potentially progress to full-blown cancer under favorable conditions.

Recent studies estimate the rate of mutation accumulation to range from 13 to 44 mutations per genome per year. This rate increases with exposure to environmental factors that induce DNA abnormalities, such as radiation, smoking, pollutants, and alcohol. Endogenous factors, including abnormalities in DNA repair mechanisms, hormonal imbalances, and oxidative stress, also contribute to mutation accumulation. Research has shown that even normal tissues can harbor clonal expansions of cells carrying carcinogenic driver mutations. These clonal expansions increase with age and exposure to both exogenous and endogenous mutagenic factors. Certain mutated cell clones are more likely to progress to cancer, depending on specific cancer driver mutations within the clone. Beyond cancer, these mutated cell clones can also contribute to other diseases, including inflammatory and cardiovascular conditions.

DNA damage can result from endogenous factors, extrinsic factors, or a combination of both. Endogenous factors include normal metabolic processes, such as oxidative stress, while extrinsic factors encompass environmental exposures, such as radiation and toxins.

Endogenous DNA damage: Endogenous factors usually result from spontaneous disruptions in normal metabolic processes, such as defects in DNA replication and disruptions in cellular signaling, leading to errors. These factors often originate from hereditary predispositions. Common forms of endogenous DNA damage include oxidative and methylation reactions occurring naturally within cells. In some cases, endogenous factors may

interact with external environmental influences, as seen in conditions such as xeroderma pigmentosum, which can further increase the risk of cancer.

Normal cellular processes, such as the Fenton reaction, combine hydrogen peroxide (H_2O_2) with iron to generate highly reactive hydroxyl (•OH) free radicals, which can cause severe DNA damage and lead to significant mutations. Additionally, DNA damage can occur internally through methylation, typically mediated by physiological methylation reactions facilitated by methyltransferase enzymes that utilize S-adenosylmethionine as a methyl donor. This process results in the formation of over 4600 methylated guanine and adenine residues, which can hinder DNA repair and polymerization, as well as disrupt the normal regulation of transcription.

Exogenous DNA damage: External factors such as chemicals, toxins, radiation, and environmental hazards are common sources of DNA damage and are largely preventable causes of carcinogenesis. Chemical combinations found in tobacco smoke, industrial processes, and chemotherapy drugs induce DNA aberrations through alkylation. Alkylating agents attach an alkyl group to the guanine base of DNA, disrupting normal protein synthesis and causing mutations. Tobacco smoke, fossil fuels, industrial chemicals such as pesticides, and high-temperature cooking can also produce aromatic amines. These amines can be converted into alkylating agents, esters, and sulfates, which target guanine bases in DNA, resulting in damage that may initiate cancer.

Recent studies indicate that environmental pollution is a significant exogenous source of DNA damage. High levels of air pollution, particularly particulate matter, have been strongly linked to an increased incidence of lung cancer. Additionally, the long-term effects of extreme temperatures resulting from global warming contribute to DNA damage. Environmental pollution now ranks as the second most common cause of cancer, surpassed only by tobacco smoke.

Since 2007, lung cancer deaths have increased by approximately 30%, a trend likely attributed to rising environmental pollution, particularly given the substantial decline in tobacco smoking rates. Particulate matter with a diameter of 2.5 µm, consisting of solid and liquid mixtures of gases, is particularly concerning. When inhaled, these particles penetrate deep into the terminal bronchioles, where they are absorbed, triggering both local and systemic inflammation. Chronic inflammation can induce significant DNA damage through the production of reactive oxygen species and free radicals.

Epigenetics

Epigenetics is defined as changes in DNA expression that are not controlled by inherited factors. This process is characterized by strict regulation and a degree of plasticity, allowing cells to differentiate. Epigenetic regulation involves the following 6 key processes:

- Posttranslational histone modifications and histone variants
- DNA methylation and demethylation
- ATP-dependent chromatin remodeling complexes
- Polycomb (PcG) and Trithorax (TrxG) complexes
- Noncoding RNAs and nuclear dynamics
- 3-Dimensional genome organization

Hyperplasticity of epigenetic regulation refers to a deviation from strict genomic control, which is characterized by a baseline plasticity that preserves cellular identity. This shift can result in the upregulation of oncogenes or the downregulation of tumor suppressor genes. Epigenetic modifications are crucial in carcinogenesis, with several key alterations frequently observed in cancer cells. These changes can arise from both genetic and nongenetic sources, including environmental, metabolic, and developmental factors such as aging.

Genetic causes include mutations in enzymes that regulate epigenetic processes, such as DNA methylases. These mutations can affect the expression of multiple genes, potentially leading to cancer initiation and progression. Similar to genetic mutations, epimutations can influence genes involved in establishing cancer hallmarks. Key epigenetic changes include global DNA hypomethylation, which results in genomic instability and oncogene activation, and promoter hypermethylation of tumor suppressor genes, effectively silencing their protective roles.

Cancer cells also exhibit altered histone modifications, including changes in acetylation and methylation, which affect chromatin structure and gene expression. Additionally, the dysregulation of microRNAs—essential for posttranscriptional gene regulation—is common in many cancers. Mutations or changes in the expression of enzymes that regulate DNA methylation and histone modifications further contribute to these epigenetic alterations. Collectively, these changes can interact with genetic mutations to initiate and drive cancer progression, making them attractive therapeutic targets due to their potentially reversible nature.

Morphology

Morphologically, a cell undergoing initiation in the carcinogenesis process displays distinct microscopic changes. The initiated cell often appears irregular, with a significant increase in the nuclear-to-cytoplasmic ratio (N/C ratio), indicating that the nucleus occupies a larger volume than the cytoplasm. This N/C ratio, also referred to as the karyoplasmic ratio, has become a crucial diagnostic parameter for cancer, as it is closely associated with cancer progression and the potential for metastasis.

Mechanisms

Carcinogenesis is a multistep process involving molecular and cellular events at both the genetic and epigenetic levels, driven by the interplay of environmental, genetic, and metabolic factors. Regardless of the specific sequence of genetic or epigenetic alterations involved, this process is typically categorized into 3 main stages—initiation, promotion, and progression.

Initiation

Cancer initiation occurs when mutations or epigenetic alterations accumulate, giving a cell a growth or survival advantage over surrounding tissues. These changes allow the initiated cell to proliferate more rapidly or evade normal regulatory mechanisms, thereby setting the stage the groundwork for tumor development.

This mutation can be hereditary or acquired. If repair mechanisms or apoptosis fail to correct the mutation after a driver gene mutation, an initiated clone of cells emerges,

signaling the onset of the initiation phase with irreparable DNA damage. Initiation confers proliferative capabilities to the cell, which may or may not progress to the promotion stage, depending on factors such as the dose and duration of exposure to an initiator. Notably, initiation is a reversible stage that can be corrected by removing the initiator. Understanding initiation is, therefore, crucial for both prophylactic and therapeutic interventions.

Upon exposure to chemical, physical, or biological factors, a cell can undergo irreversible genetic changes, potentially evolving into a neoplastic cell clone. This transformation occurs rapidly, within minutes to hours. The initiated cell can remain in this state indefinitely without causing harm or being detected by the organism's defense systems, as it does not exhibit any noticeable phenotypic changes. Sporadic mutations, which account for approximately 75% of colorectal tumors, are particularly prevalent in this stage.

The initiation of cancer involves critical factors such as DNA damage, genetic mutations, and inflammation. Notably, there may be considerable overlap in the mechanisms through which genetic alteration, DNA damage, and inflammation contribute to the initiation process. The mutated cell, known as an initiated cell or initiated clone, subsequently undergoes clonal expansion. During this phase, the mutated cell and its progeny replicate, forming a population of cells that share identical genetic alterations.

After initiation, carcinogenesis advances through the promotion and progression stages. Understanding the mechanisms involved in the initiation phase is crucial for developing strategies to prevent, detect, and treat cancer. Identifying specific carcinogens and recognizing the genetic alterations associated with initiation have led to significant progress in cancer research, paving the way for more targeted therapies.

Recent research suggests that cell initiation may result from epigenetic changes rather than genetic mutations. Environmental factors, such as exposure to non-mutagenic carcinogens, can trigger these epigenetic events. These modifications may serve as the initial alteration in cell properties, potentially leading to the development of the first hallmark of cancer. Alternatively, these epigenetic changes may prime cells to respond more readily to a subsequent genetic mutation. Conversely, an epigenetic mutation may arise from a mutation in genes that regulate epigenetic processes, such as DNA methyltransferase genes *DNMT1* and *DNMT3A*.

Promotion and Progression

Promotion involves the expansion of initiated cells into a detectable tumor, whereas progression entails the acquisition of additional genetic alterations that increase tumor aggressiveness. Progression can also include the development of metastatic capabilities, enabling the tumor to spread to distant sites in the body.

Promotion refers to the stage of cancer development where altered or damaged cells, having evaded apoptosis, become immortal and acquire mutations that support their proliferation and survival from a single progenitor cell. Although still considered premalignant and reversible, the growth and sustained survival of these transformed cells are stimulated, eventually leading to the formation of a discernible tumor.

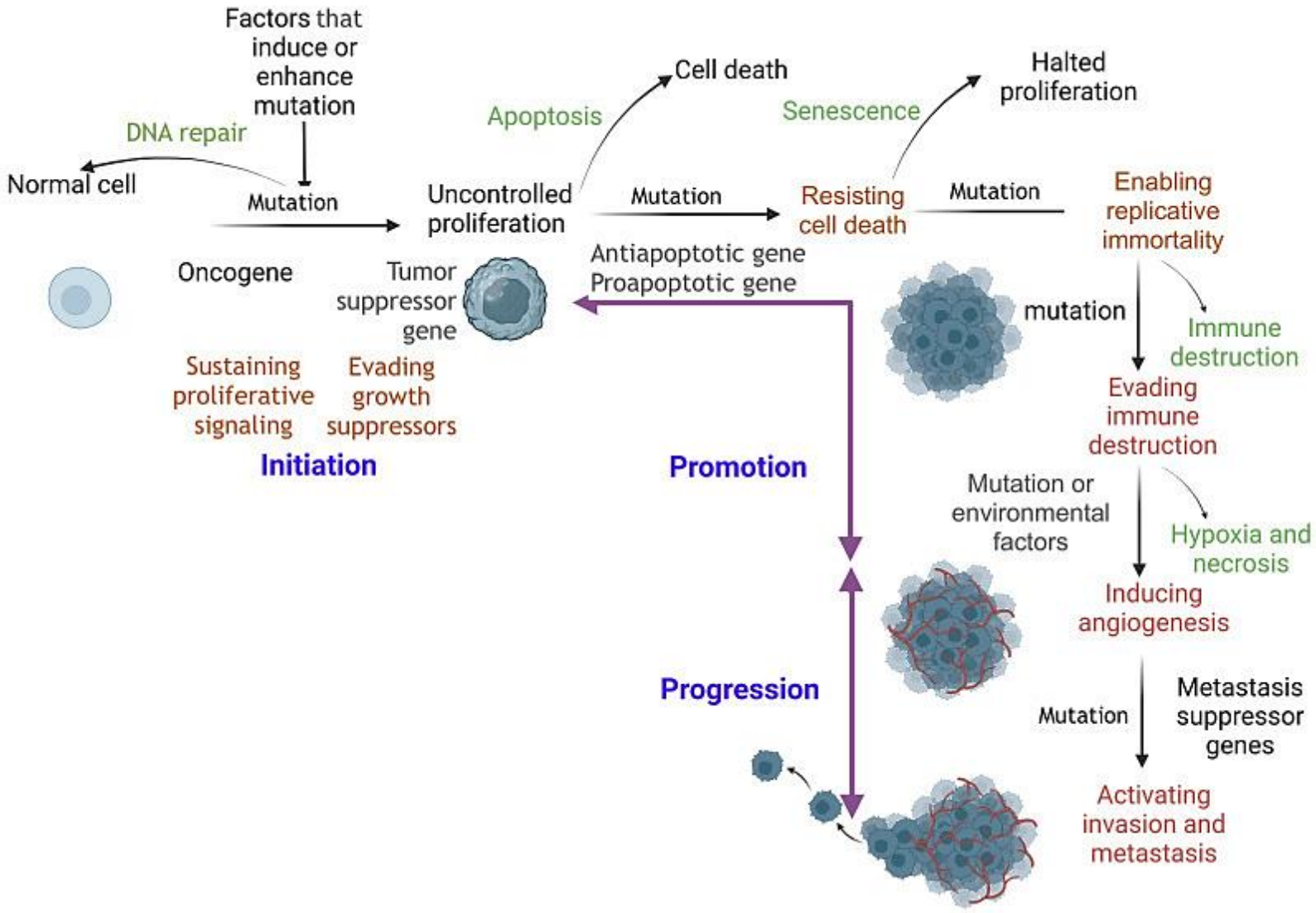
This intricate process involves various intrinsic and extrinsic factors, including hormones, growth factors such as transforming growth factor-β (TGF-β), and chronic exposure to certain chemicals or irritants, all of which shape the tumor microenvironment. These agents facilitate carcinogenesis by promoting the growth of initiated cells. The promotion stage depends on the duration and intensity of exposure to these promoting agents, which can accelerate the progression of initiated cells toward distant tissue invasion.

Some events may overlap between the promotion and progression stages. Hormones, such as estrogen, can act as stimulatory agents in certain cancers. For instance, estrogen promotes breast tissue growth and is associated with both the promotion and progression of specific breast cancers. During the promotion stage, cellular proliferation is stimulated, particularly in cells with acquired genetic alterations that drive uncontrolled growth and increased responsiveness to these signals.

Epigenetic modifications, such as alterations in DNA methylation patterns and histone modifications, can promote cancer development by influencing the expression of genes that regulate the cell cycle and apoptosis, ultimately affecting cell behavior. Furthermore, chronic inflammation can facilitate cancer progression by creating a microenvironment rich in growth-promoting signals and factors that enhance cell survival.

Inflammatory cells release cytokines and growth factors that promote the survival and proliferation of cancer cells. Angiogenesis, the formation of new blood vessels, is often associated with tumor growth during the promotion stage. Tumors depend on an adequate blood supply for their growth and development, and angiogenic factors secreted by tumor cells or adjacent stromal cells stimulate blood vessel formation within the tumor. The tumor microenvironment—including immune cells, fibroblasts, and extracellular matrix components—supports tumor growth. The interactions and cooperation between tumor cells and their surrounding environment are essential for the survival and proliferation of cancer cells.

Understanding the progression stage is crucial for developing interventions to halt the carcinogenic process. By targeting factors that promote cancer cell growth and disrupting the pathways that support them, it may be possible to impede or slow the transformation of precancerous cells into malignancies. Additionally, identifying and addressing modifiable risk factors that contribute to cancer progression is essential for effective cancer prevention and control. In the final stage of carcinogenesis, cells that have undergone initiation and promotion acquire the necessary mutations to invade surrounding tissues, marking the onset of invasive cancer.


Factors that induce or enhance mutation
DNA repair
Normal cell
Mutation
Uncontrolled proliferation
Apoptosis
Cell death
Mutation
Resisting cell death
Senescence
Halted proliferation
Mutation
Enabling replicative immortality
Oncogene
Tumor suppressor gene
Antiapoptotic gene
Proapoptotic gene
Sustaining proliferative signaling
Evading growth suppressors
Initiation
Promotion
mutation
Immune destruction
Evading immune destruction
Mutation or environmental factors
Hypoxia and necrosis
Inducing angiogenesis
Progression
Mutation
Metastasis suppressor genes
Activating invasion and metastasis